

Delocalized aromatic molecules with matched electron-donating and electron-withdrawing groups enhancing insulating performance of polyethylene blends

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Abstract

Seven delocalized aromatic molecules with electron-donating and electron-withdrawing groups are applied as voltage stabilizers to improve the insulation properties of polyethylene blends. Voltage stabilizers 1 wt% are added into the blends (90 wt% low-density polyethylene and 10 wt% high density polyethylene) by diffusion loading method. Electrical measurements including electrical treeing, space charge distribution, and direct current conductivity are conducted to disclose their effects. The results show that the co-existence of matched electron-donating and electron-withdrawing groups in the molecules is favorable for the insulation properties. A 50% increase of tree initiation voltage is achieved with the addition of 3-aminobenzoic acid, which is also able to inhibit the space charge and decrease the conductivity at lower temperature. Besides, the grafting sample of the optimal molecule is successfully prepared and displayed enhanced electrical properties. Finally, the common internal mechanism of the delocalized molecules on different electrical properties is revealed.

1 | INTRODUCTION

Nowadays, power network is in high-speed development, for example, the renewable solar and wind energy have

served as power source increasingly, which should resort to the technology of long-distance and large-capability electric power transmission. The polymer insulating cables have gradually substituted the oil-filled cables and mass impregnated cables due to the excellent features, such as light weight, prominent insulation, good flexibility, and low cost.^[1–3] Cross-linking polyethylene (XLPE)

Abbreviations: LH, 90% of LDPE and 10% HDPE; TIV, tree initiation voltage.

has been regarded as the most common high-voltage power cable insulation, by which the power transmission at 640 kV is achieved.^[4,5] Furthermore, XLPE possesses high breakdown strength, low dielectric loss property, and thermo-mechanical stability, which can significantly enhance the performance of insulation. However, its cross-linked network and thermoset makes it nonrecyclable.^[6,7] Alternatively, thermoplastic power cable insulation, such as polypropylene (PP), low-density polyethylene (LDPE), and high-density polyethylene (HDPE), has received growing attention for its recyclability and environment-friendly processing.^[8]

However, some crucial issues for the thermoplastic insulation still need to be solved. Electrical treeing, also named as pre-breakdown degradation, and space charge accumulation in high voltage direct current (HVDC) transmission may cause an insulation failure.^[3,9–12] Generally, three stages are involved in the process of electrical treeing: initiation, propagation, and breakdown.^[13] It has been verified that the initiation is always accompanied by charge injection for all transmission of alternating current (AC) voltage, direct current (DC), voltage or impulse voltage,^[14] while the mechanism of the energy transfer from injected charge to the polymer has not been determined. Among them, electroluminescence photodegradation theory, charge injection and extraction theory, and trap theory are the most widely used.^[15–18] Although these three theories emphasize on different pathways, the vital effect of hot electron is generally acceptable by most researchers. As a consequence, the key to suppressing the initiation of electrical treeing is to reduce the generation of hot electrons and dissipate their energy in order to prevent them from reaching harmful energy levels and avoid their attack on the polyethylene chain.^[19] So far, as the generation of hot electrons is not easy to control, more concentration has been devoted to the latter. The space charge accumulation in HVDC transmission can be caused by the by-products of the manufacturing process and the external strains, which may result in distortion of electric field, generation of DC electrical tree, and even the final breakdown.^[3,12,20–22] It is highly applied that deep traps are introduced to the insulation polymer to capture the injected charges and obstruct the mobility of the charge carrier and consequently to hinder the space charge accumulation.^[11,22,23]

Several strategies, including nanoparticles introducing, polymer blending, voltage stabilizer adding, and chemical grafting, are commonly reported to inhibit electrical treeing and space charge accumulation of insulation materials. MgO, Al₂O₃, SiO₂, ZnO, BN, fullerene, and graphene are popularly applied nanoparticles.^[23–29] It is reported that the introduction of nanoparticles can introduce deep traps into insulation polymer, which accordingly suppresses the space

charge accumulation, decreases the conductivity, and increases the breakdown strength. However, the nanoparticles tend to migrate and agglomerate; therefore, the dielectric properties of insulation materials can be enhanced obviously only under the premise that the nanoparticles are well dispersed in the polymers.^[24,30] Two or more polymer blends are also frequently used, which can increase the thermo-mechanical performance and change the crystallization and morphology of insulation polymers.^[2,31,32] Voltage stabilizer adding, a pretty popular strategy to improve the dielectric properties of insulation, can be traced back to 1960s in 20 centuries. In recent years, many chemicals have served as effective voltage stabilizers, such as aromatic ketone derivatives,^[10,11] benzil derivatives,^[33,34] thioxanthone derivatives,^[35] and polycyclic aromatic hydrocarbons.^[36,37] As one can see, the common character of the reported voltage stabilizers is the widespread existence of aromatic nucleus and carbonyl group in their structures. Aromatic nucleus possesses high electron affinity and low ionization energy for its typical conjugated and π electron-delocalized structure,^[10,38] which can accept the hot electron attacking prior to the polymer segments and then dissipate the energy to release unharmed hot electrons. On the other hand, as a polar group, carbonyl group affords deep trap which could reduce the charge mobility and raise the charge injection barrier, leading to the suppression of space charge accumulation. In addition, carbonyl group can also intensify electron affinity of the voltage stabilizer as it is an electron-withdrawing group. Chemical grafting gradually comes into our sight due to the migration and inferior compatibility with polymer of voltage stabilizers. Maleic anhydride is the most used grafted chemical^[3,11,21,23] and acrylic acid derivatives,^[39] siloxane derivatives^[40] as well as 1-(4-vinyloxy)phenylethanone^[41] are also applied. Nonetheless, the complex grafting progress and the uncontrollable grafting ratio limit the application of the strategy.

It is well known that aromatic nucleus attached with carbonyl group is an effective structure for voltage stabilizers. However, an original idea occurs to us that compared with the isolate existence of electron-withdrawing groups on aromatic nucleus, the additional introduction of a matched electron donating group may be more favorable. This structure seems like a micro electric field, in which the aromatic nucleus of a conjugated π electron-delocalized structure enables the formation of electronic channel, and the electron withdrawing groups combined with the electron donating groups form two electric poles. Therefore, the hot electrons can oscillate to achieve enough buffer and more thorough dissipation of energy. In the meantime, the intermolecular hydrogen bond also promotes the hot electron buffer,^[42] as it may connect among these micro electric field structures to obtain a giant electron buffer system. There is some available

reported data of experiment supporting our idea. Jarvid et al.^[33] presented that the tree initiation voltage (TIV) of benzil with methoxyl, methyl ester and dimethyl amino side-chains (as 3b, 3c, and 3e in Figure 1) increased by 43.6, 72.0, and 73.3% in contrast with pure XLPE, indicating the positive effect of the co-existence of withdrawing and donating groups. However, vast density functional theory (DFT) calculations have been devoted to reveal the selection criteria of voltage stabilizers. As reported, the HOMO–LUMO energy gap (E_g) of molecule calculated by DFT can act as a measurement of the stabilizer performance. A smaller E_g of the voltage stabilizer is more favorable because it means that the molecules are more likely to be activated.^[10,19,38,43] It is concluded that introduction of a suitable electron donating group can decrease the E_g obviously compared with the aromatic nucleus only consisting of electron-withdrawing substituents (Figure 1).^[19,34,43] Additionally, the introduction of another electron withdrawing group may also be beneficial for decreasing E_g . However, Yang et al.^[43] has reported that the introduction of more than one electron-withdrawing groups brought about larger gap between E_t (h) and E_t (e) while the introduction of electron-donating group to benzil caused smaller gap between trap energy E_t (h) and trap depth E_t (e), which is essential for the suppression of space charge accumulation.

Herein, a series of molecule additives was applied as voltage stabilizers to demonstrate our speculation and the electrical properties of the composites were investigated to test the effect of the voltage stabilizers. In addition, a selection criterion of voltage stabilizers was also proposed.

2 | EXPERIMENTAL

2.1 | Materials

The base stocks, LDPE and HDPE pellets, are from Nanjing Zhongchao New Materials corporation (China).

Dicumyl peroxide (DCP), 3-aminophenylboronic acid, and 3-aminobenzoic acid are purchased from Shanghai Aladdin Bio-Chem Technology Co., Ltd (China). 4-Cyanophenylboronic acid, 3-vinylphenylboronic acid, 4-propoxyphenylboronic acid, and 4-(dimethylamino)phenylboronic acid are purchased from Shanghai Macklin Biochemical Co., Ltd (China). 2-Methoxy-5-pyridineboronic acid is purchased from Adamas-Beta Co., Ltd (China). The applied solvents methanol, acetone, and xylene are purchased from Sinopharm Group Co., Ltd (China).

2.2 | Sample preparation

For diffusion loading method, 54 g LDPE and 6 g HDPE pellets were frozen in liquid nitrogen, ground (Retsch ZM200), and impregnated with equal volume of certain solvent (methanol was used for mixing with voltage stabilizers of 3-aminophenylboronic acid, 4-cyanophenylboronic acid, 3-vinylphenylboronic acid, 4-propoxyphenylboronic acid, 2-methoxy-5-pyridineboronic and 4-(dimethylamino)phenylboronic acid; acetone was used for mixing with voltage stabilizer of 3-aminobenzoic acid) in a round-bottom flask. Voltage stabilizer of 0.6 g (1 wt% of the polymer matrix) was first predissolved in 5 ml of the corresponding solvent and introduced to the round-bottom flask. The mixture was stirred at room temperature for 2 hr, rotationally evaporated, and finally vacuum dried at 60°C for 12 hr. The dried mixture was hot pressed in the plate vulcanizing machine at 160°C and 20 MPa for 180 s, followed by quenching with cold water at constant pressure.

3-Aminobenzoic acid was chosen to graft on the blends of LDPE and HDPE by solution grafting method through the mechanism of free radical grafting (grafting structure as shown in Figure S2). 54 g LDPE, 6 g HDPE, 0.6 g additive (1 wt% of the polymer matrix), and 250 ml xylene were added into the a 1,000-ml three-necked

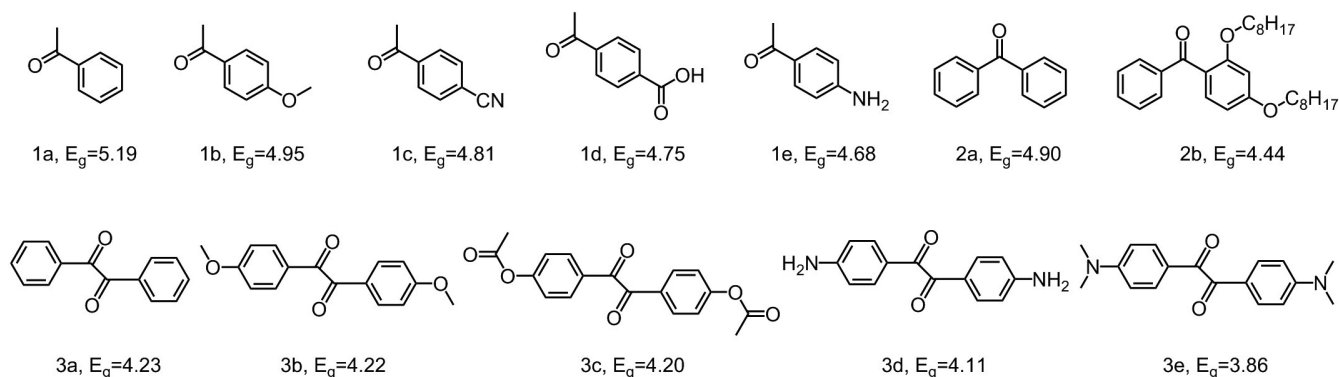


FIGURE 1 Reported E_g of voltage stabilizers calculated by density functional theory calculations: 1a, 1b, 1c, 1d, and 1e are from Reference^[43]; 2a and 2b are from Reference^[19]; 3a, 3b, 3c, 3d, and 3e are from Reference^[34]

round-bottomed flask and heated to 140°C with intensive mechanical stirring. The same moles of DCP as the additive were dissolved in 50 ml xylene, which was dropwise added into the above flask after the complete dissolution of the polymer. The grafting reaction lasted for 2.5 hr after the dropping completion. The whole process was conducted under the nitrogen protection. In order to remove the decomposition product of DCP and ungrafted 3-aminobenzoic acid, the mixture was washed in acetone for four times and further vacuum dried at 100°C for 12 hr. The hot pressed process is the same as the above.

2.3 | Chemical characterization and electrical property test

Please see also the details in our previous publications.^[44,45]

Chemical characterization: The Fourier transform infrared spectroscopy (FTIR) spectra of all the samples were got on a Nicolet iS50 FTIR spectrometer (Thermo Nicolet Ltd., USA) with the wavelength range of 4,000–500 cm^{-1} . The melting point and degree of crystallinity of each LDPE composites were determined by Differential Scanning Calorimetry (DSC, Q200, TA Instruments Inc., USA). Each of the shredded samples (5–6 mg) was loaded in a platinum crucible with a cap and tightly compressed by a tablet press. After that, a heating–cooling–heating operation was performed to the above samples under the nitrogen flow rate of 50.0 ml min^{-1} . First, the samples were heated from the room temperature to 160°C at a rate of 10°C min^{-1} in order to eliminate the thermal histories. After maintaining the temperature for 5 min, it was decreased to 40°C at a cooling rate of 10°C min^{-1} and kept for another 5 min. Finally, another heating operation was conducted to 160°C at a heating rate of 10°C min^{-1} . The melting temperature (T_m) and degree of crystallinity (X_c) of LDPE were determined by the first peak in the melting endotherm of the final scan. Thermogravimetric analysis (TGA) was performed using a Pyris 1 TGA instrument (Perkin-Elmer Inc., USA). About 3 mg of each shredded sample was placed in a platinum crucible and heated from 50 to 600°C at a heating rate of 10°C min^{-1} under a nitrogen atmosphere.

Electrical property test: The needle-plane type geometry was applied in the electrical treeing experiments, as displayed in Figure S1: the high voltage electrode is a steel needle with the shank diameter of 1 mm and tip radius of $5 \pm 1 \mu\text{m}$; the dimension of the sample is $30 \times 15 \times 2 \text{ mm}$; the steel needle tip is embedded $2 \pm 0.2 \text{ mm}$ from the plane electrode. The test sample was immersed into silicone oil to avoid the partial discharge. The needle electrode was connected to a HVAC source ramping at about 80 V s^{-1} and the other side of

the sample was grounded. A partial discharge monitoring system and a real-time microscopic imaging system were used to detect the electrical tree initiation and propagation. For each composite, 10 samples were measured and the treeing inception voltage was analyzed via the two-parameter Weibull distribution. The DC conductivity test was conducted by using the three-electrode system, in which the high voltage electrode is a stainless steel cylinder with a diameter of 45 mm and the measuring electrode was 30 mm in diameter, while the guard ring with diameter of 45 mm allowing for eliminating surface leakage current. The samples with a dimension of $\Phi 9 \text{ cm} \times 185 \mu\text{m}$ were tested under the DC electric field of 10, 20, and 30 kV mm^{-1} for 20 min, and then the depolarization process was conducted for 5 min. The test temperatures were 30, 40, 50, and 60°C under every field. In addition, the experiment was repeated for twice and the average was taken. The space charge behavior of the samples was tested by a pulsed electro-acoustic method with a pulse voltage of 100 Hz and a pulse amplitude of 10 ns. For the test structure, a semiconductive material with a diameter of 2 cm was sandwiched between the high-voltage electrode and the test sample with a dimension of $\Phi 9 \text{ cm} \times 185 \mu\text{m}$. Besides, the ground aluminum electrode is at the bottom. The test process included a polarization process with DC electric field of 50 kV mm^{-1} for 30 min at room temperature and a followed depolarization process for 10 min.

3 | RESULTS AND DISCUSSION

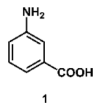
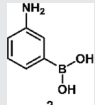
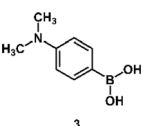
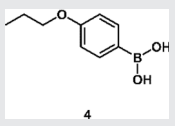
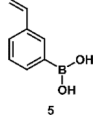
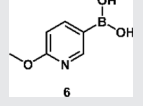
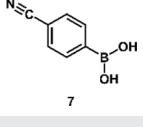
3.1 | Analysis of the samples prepared by diffusion loading method

Seven small molecules containing electron-donating or withdrawing functional groups, that is, 3-aminobenzoic acid (denoted as 1), 3-aminophenylboronic acid (2), 4-(Dimethylamino)phenylboronic acid (3), 4-propoxy phenylboronic acid (4), 3-vinylphenylboronic acid (5), 2-methoxy-5-pyridineboronic acid (6), and 4-cyanophenyl boronic acid (7), were designed to mix with 90% of LDPE and 10% HDPE (denoted as LH) as voltage stabilizers. The L/H ratio followed our previous work as it proved to have excellent electrical performance. The TIV, DC conductivity, and space charge were detected to evaluate their insulation properties.

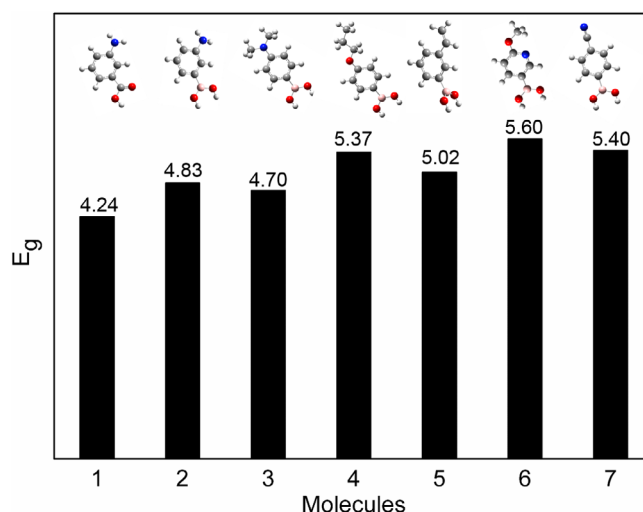
3.1.1 | Electrical tree tests

Table 1 displays the TIV data of the samples, provided by a statistical analysis, Weibull distribution (Figure S3a).

TABLE 1 Scale parameters of the two-parameter Weibull distribution for the polyethylene blends with/without voltage stabilizers

Sample name	Voltage stabilizer	TIV (kV)	Increment ratio/LH (%)
LH + 1		23.8	50.3
LH + 2		22.7	43.1
LH + 3		20.1	26.6
LH + 4		17.6	11.0
LH + 5		17.1	8.0
LH + 6		19.7	24.6
LH + 7		15.1	-4.8
LH	—	15.8	—

The sample LH + 1 possesses the highest TIV of 23.8 kV among all the tested voltage stabilizers, which enhances by more than 50% with respect to the 15.8 kV TIV of the reference LH blends. This result meets quite well with our understanding that the co-existence of withdrawing and donating groups in a benzene ring would improve its antivoltage properties. $-\text{NH}_2$ and $-\text{COOH}$ in molecule 1 are generally viewed as the strong donating and withdrawing substitute groups, respectively. As a result, their combination with benzene ring may form a micro electric field to resist the exogenous high energy electrons. The voltage stabilizers 2–5 contain an electron-withdrawing group of $-\text{B}(\text{OH})_2$ in each molecule. The second substitute group is $-\text{NH}_2$, $-\text{CH}=\text{CH}_2$, $-\text{OCH}_2\text{CH}_2\text{CH}_3$, or $-\text{N}(\text{CH}_3)_2$, all of which are viewed as electron-donating groups. Consequently, all of the four voltage stabilizers exhibit positive effect on enhancing the TIVs by 8–43%

**FIGURE 2** The energy gap E_g of the designed molecules [Color figure can be viewed at wileyonlinelibrary.com]

with respect to that of the LH blends. However, the molecule 7 only with two electron-withdrawing groups, cyano and boric acid group, displays a negative effect, which decreases the TIV by about 5%. Although the electron-withdrawing groups can also serve as electron traps, the co-existence of matched electron-donating and electron-withdrawing groups in the aromatic molecules will achieve enough buffer of the hot electrons and more thorough dissipation of their energy. This can be ascribed to the reason for the inferior performance of molecule 7. In addition, the positive effect of the voltage stabilizer 6 demonstrates that other aromatic nucleus such as pyridine ring may be also effective on the TIV increase.

3.1.2 | DFT calculations

In order to quantitatively determine the electronic donor-acceptance capability of the tested voltage stabilizers, we further calculated the energies of HOMO and LUMO using B3LYP functional with 6-311 + G(d, p) basis set by DFT method (Figure 2). The energy gaps (E_g) between the highest occupied orbitals and the lowest unoccupied orbitals are also displayed. According to Koopman's theorem,^[10,34] a molecule with lower LUMO has larger electron affinity energy, indicating its greater acceptance of hot electrons; while a molecule with higher HOMO represents smaller ionization potential, indicating it is easier to be dissociated. These two properties are beneficial for a voltage stabilizer to prevent the polymer matrix from degradation. Meanwhile, the HOMO level and the LUMO level are also severally related to the electron-donating and electron-withdrawing ability of a molecule.^[46,47] In terms of LUMO level, the electron-

withdrawing ability of the functional groups is disclosed that $-\text{CN}$ (-2.05 eV) $>$ $-\text{COOH}$ (-1.54 eV) $>$ $-\text{B}(\text{OH})_2$ (-0.82 eV); while in terms of HOMO level, the electron-donating ability of the functional groups is disclosed that $-\text{N}(\text{CH}_3)_2$ (-5.33 eV) $>$ $-\text{NH}_2$ (-5.64 eV) $>$ $-\text{OCH}_2\text{CH}_2\text{CH}_3$ (-6.11 eV) $>$ $\text{CH}=\text{CH}_2$ (-6.23 eV). As shown in Figure 2, the molecule 1 with $-\text{NH}_2$ and $-\text{COOH}$ has the smallest E_g (4.24 eV); the molecule 2 with $-\text{NH}_2$ and $-\text{B}(\text{OH})_2$ and 3 with $-\text{NH}_2$ and $-\text{N}(\text{CH}_3)_2$ also possess rather smaller E_g (4.83 and 4.70 eV). In contrast, the molecule 7 with two electron-withdrawing groups $-\text{CN}$ and $-\text{B}(\text{OH})_2$ has a very large E_g (5.40 eV). We find that the groups $-\text{NH}_2$ and $-\text{COOH}$ possess strong electron-donating and electron-withdrawing ability, respectively. Consequently, apart from the introduction of polar groups and the formation of stronger conjugation in the molecule,^[19,34] we speculate that the co-existence of matched electron-donating and electron-withdrawing groups also contributes to a smaller E_g , which conforms to the reference that the combination of exceptionally potent π -donor and π -acceptor groups leads to very small HOMO–LUMO gaps.^[48] In order to disclose the relationship between E_g and TIV, the linear fitting was conducted for the two parameters. An evident trend was depicted in Figure 3 that E_g decreases with the TIV increasing. To be specific, voltage stabilizer 1 was characterized of the smallest E_g and the highest TIV accordingly. On the other hand, voltage stabilizer 7 possesses a large E_g and a low TIV accordingly. Moreover, the trend is basically tenable for voltage stabilizers 2, 3, 4, and 5. However, voltage stabilizer 6, with a pretty large E_g , shows a high TIV, which may be ascribed to the electron-deficient pyridine ring. It is

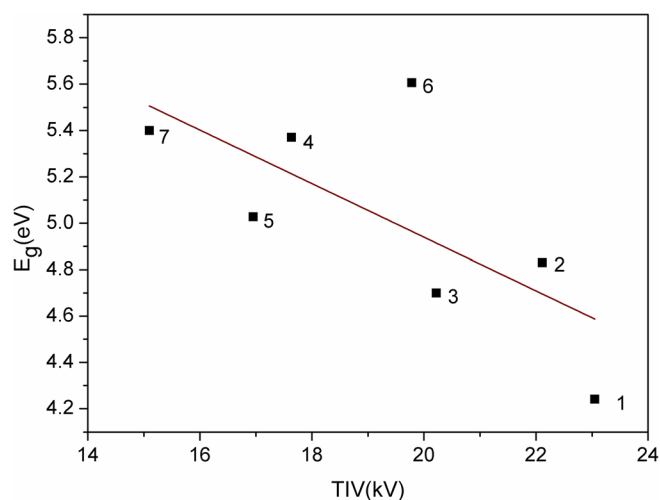


FIGURE 3 The linear fitting result between E_g and tree initiation voltage of the samples [Color figure can be viewed at wileyonlinelibrary.com]

found that the coefficient of determination R^2 is 0.3799 with the data of stabilizer 6 and 0.9815 without the data.

3.1.3 | DC conductivity tests

The DC conductivity of the samples is depicted in Figure 4, from which we can see that it increases obviously with the temperature and electrical field enhancing. The two factors play a similar role in the hopping conduction—the mobility and the release rate from traps of the carriers increases when the temperature and electrical field are enhanced, leading to the increment of conductivity.^[49–51] With the voltage stabilizers adding, the conductivity rises in varying degrees except for the addition of 1 and 2, which decrease the conductivity at lower temperatures of 303 and 313 K (inset graphs in Figure 4). Especially, when adding stabilizer 1, the conductivity also decreases evidently at 323 K but increases at 333 K. Connecting with the TIVs of the samples, we find that the stabilizers show consistency in the effect on TIV and conductivity. That is to say, stabilizers who performed well in the TIV increasing also show a positive effect on conductivity decreasing. For example, stabilizers 1 and 2 display superior performance in both increasing TIV and decreasing conductivity but stabilizer 7 shows inferior performance in both the two tests. From the tight correlation of the experimental results, a similar essence is inferred in the mechanism of the conductivity reducing and electrical tree inhibiting, which has been elucidated as trap of the charge carriers for the former and oscillation of the hot electrons for the latter. Generally, traps come into failure when they are filled with carriers that will be re-released to the conduction band; while for the designed molecules, the electron carriers (equal to negative charges) and hole carriers (equal to positive charges) captured by electron and hole traps may neutralize just as the reported charge neutralization.^[52] This may be ascribed to the structural features of the designed molecules: the aromatic nucleus can serve as the bound area of the carriers and the electron-withdrawing and electron-donating groups may serve as electron and hole traps, respectively.^[53]

3.1.4 | Space charge tests

The dynamic space charge behavior of the samples under 50 kV/mm DC electric field is displayed in Figure 5, in which the distance between the two vertical dashed lines is equal to that from anode to cathode. There are no large number of hetero-charges near the anode and negative charges in the bulk of the reference LH. As reported, the added HDPE can serve as nuclei to facilitate the

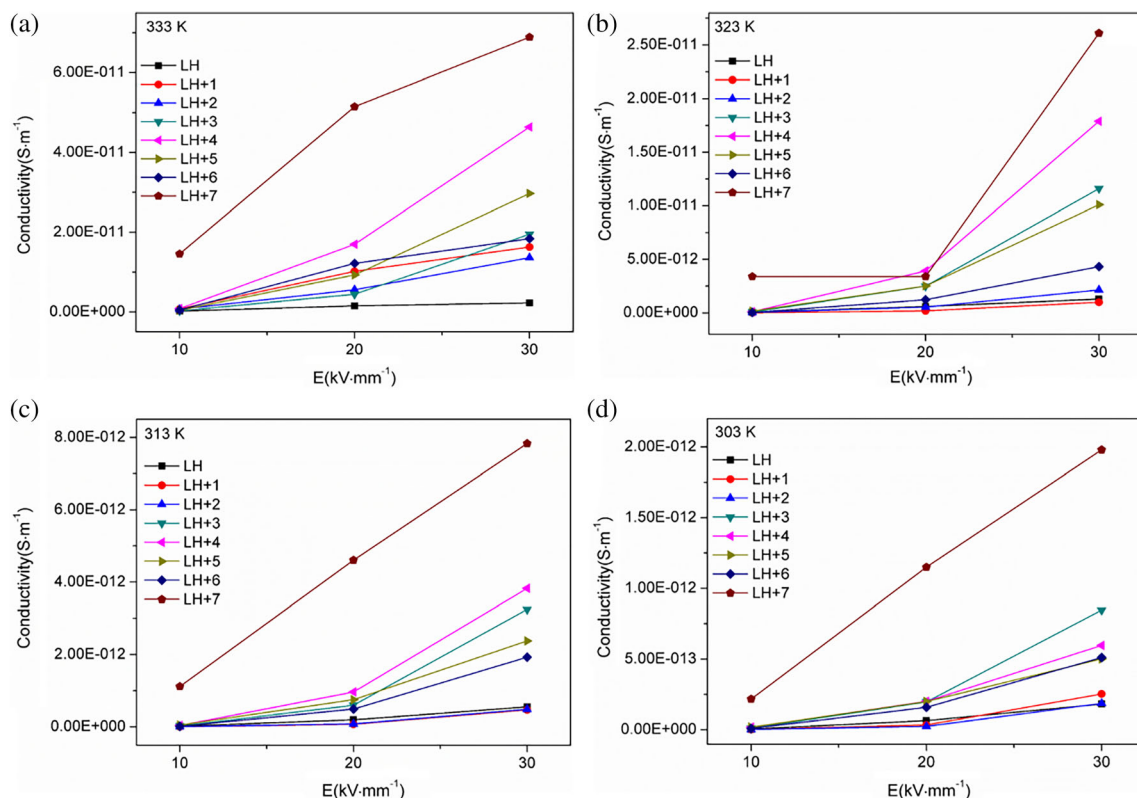


FIGURE 4 Temperature and electrical field dependence of direct current conductivity [Color figure can be viewed at wileyonlinelibrary.com]

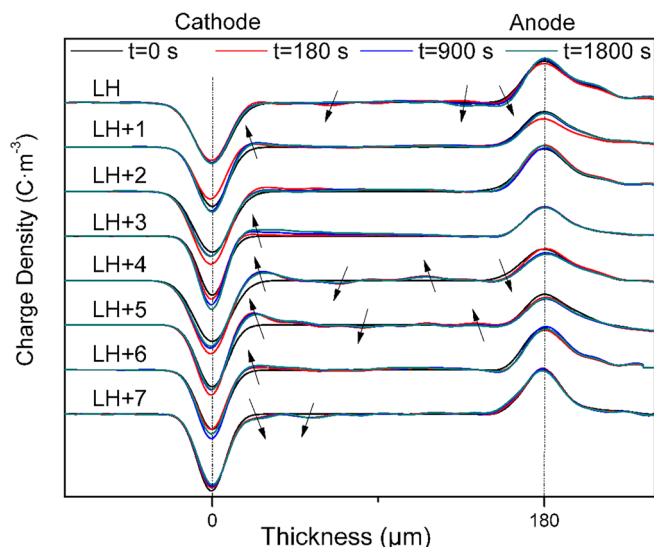


FIGURE 5 Space charge distributions of the diffusion loading samples [Color figure can be viewed at wileyonlinelibrary.com]

crystallization and decrease the average size of spherulites.^[31] Hence, more spherulite boundaries generate and provide the path for the migration of charge carriers in semicrystalline polymers.^[54] For the stabilizer-introduced samples, the hetero-charge accumulation near the cathode is ubiquitous except for 2 and 7. This can be ascribed that the introduced stabilizers are ionized to release charges

and the mobility of electrons is stronger than that of holes under high DC electric field.^[55] Notably, the sample LH + 2 shows top performance of the space charge inhibiting, in which hardly any accumulated space charges can be observed. Besides, stabilizers 1, 3, and 6 also manifest effective space charge suppression. In these samples, only a small number of positive charges are found near the cathode and no evident charge accumulation shows in the bulk. Inversely, stabilizers 4 and 5 perform badly in space charge inhibiting and rather larger number of positive and negative charges are observed in the two poles and the bulk. Particularly, the sample LH + 7 only presents negative charge accumulation in the cathode and the bulk. The difference from other stabilizers may be ascribed to the two strong electron-withdrawing groups in stabilizer 7, which tends to trap electrons. It can be seen that the similarity of the well-behaved stabilizer is the co-existence of matched electron-withdrawing and electron-donating groups on the aromatic nucleus, in which negative and positive charges can be trapped and then neutralized.

3.2 | Analysis of the grafting sample

Grafting method was adopted to avoid the migration and inferior compatibility with polymer of the voltage

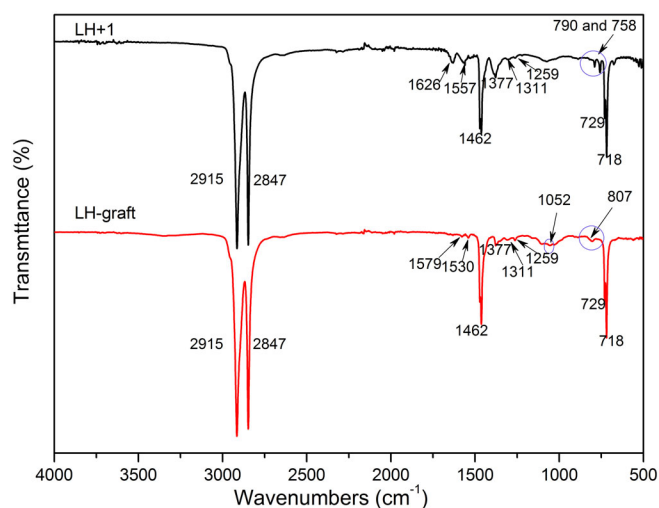


FIGURE 6 Fourier transform infrared spectra of the samples prepared by diffusion loading and grafting method [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

stabilizers. The optimal 3-aminobenzoic acid was chosen as the grafting molecule. In the meantime, physicochemical property characterizations and electrical tests were conducted to disclose the performance of the grafting sample.

3.2.1 | Characterization of the grafting sample

FTIR has been performed to determine if 3-aminobenzoic acid was grafted on the polymer chains (Figure 6). The similar peaks (2,915, 2,847, 1,462, 1,377, 729, and 718 cm^{-1}) can be observed in the two curves of diffusion loading sample (LH + 1) and grafting sample (LH-graft), which are assigned to CH_2 and CH_3 groups in PE (details in Table S1).^[56,57] Besides, the peaks at 1,626 and 1,557 cm^{-1} in LH + 1 as well as 1,679 and 1,530 cm^{-1} in LH-graft belong to $\text{C}=\text{C}$ framework vibration.^[58] It is noteworthy that the peaks in grafting sample are weaker due to the limited grafting ratio. Probably, because the grafting operation changes the chemical conditions of the benzene ring, the blue shift is also obviously observed. The peaks at 1,311 and 1,259 cm^{-1} are ascribed to $\text{C}-\text{N}$ (aryl carbon) and $\text{C}-\text{O}$ stretching vibration.^[59,60] However, a peak of $\text{C}-\text{N}$ (alkyl carbon) 1,052 cm^{-1} arises after grafting, indicating the successful grafting.^[61] Another sign for the successful grafting is that the peaks of $\text{N}-\text{H}$ (amino) symmetric and asymmetric deformation vibration at 790 and 758 cm^{-1} disappear while the peak of $\text{N}-\text{H}$ (secondary amino) deformation vibration at 807 cm^{-1} appears accordingly.^[58,62]

The DSC (Figure S4) and TGA/DTG (Figure S5) of the samples have been characterized and the correlated parameters are displayed in Table 2. Compared with the

TABLE 2 Character temperatures and degree of crystallinity of the samples

Sample	T_m ($^{\circ}\text{C}$)	X_c (%)	T_o ($^{\circ}\text{C}$)	T_p ($^{\circ}\text{C}$)
LH	107.8	35.9	439.2	462.5
LH-graft	106.6	35.3	438.9	462.2

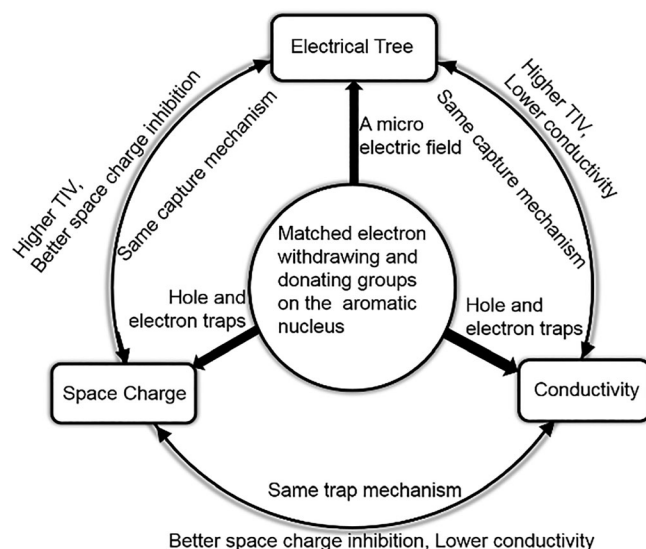


FIGURE 7 The related internal mechanism and performance of the stabilizers on the electrical properties

reference LH, the melting temperature (T_m) and the crystallinity (X_c) of the grafting sample only show a slight decrease, indicating little change in the material microstructure at the current grafting degree.^[63,64] This also reveals that the effect of the grafting molecule on the electrical properties is not caused by the polymer structure change. Both the onset decomposition temperature (T_o) and the peak temperature of maximum decomposition rate (T_p) only decrease 0.3 $^{\circ}\text{C}$ after the grafting operation, which means the operation has little influence on the thermal stability of the material.^[56]

3.2.2 | Electrical tests of the grafting sample

The grafting sample TIV of 20.03 kV is provided by Weibull distribution (Figure S3b). The data show an evident increase by about 29% compared with the LH sample (15.8 kV). The increment ratio is satisfactory although it is not as high as the corresponding diffusion loading one (23.8 kV), which may be ascribed to the limited grafting ratio. It is acknowledged that the grafting ratio is usually as low as the teens or even a few percent regardless of the grafting method.^[65,66] However, the DC

electrical properties of the grafting sample are not so consistent with the diffusion loading ones for DC conductivity and space charge (Figures S6 and S7). The grafting conduction displays a substantial decrease in the DC conductivity but no obvious inhibition in space charge. It is speculated that the two properties vary in terms of the content sensitivity. To be specific, the lower content may be more favorable for the conductivity dropping due to the intrinsic conductivity of polar molecules but cannot reach the threshold of the space charge inhibition.

4 | DISCUSSIONS

Currently, benzil and thioxanthone derivatives are the most used molecule voltage stabilizers. For example, Jarvid et al and Wutzel et al^[33,35] have reported that compared with the neat XLPE, the introduction of these two derivatives can increase the TIV about 12–73% and 15–55%, respectively. In this paper, the designed optimal molecules can achieve the topmost 50% TIV increase with respect to reference LH, which is comparable to the reported data. Besides, they are smaller or rather simpler than the reported, which may reduce the possibility of the crystalline region being disordered to inhibit the growth of electrical tree and the generation of deep traps in the bulk. Additionally, compared with the most used inorganic nanoparticles, the designed molecules have the comparative ability of space charge inhibiting and more excellent compatibility with polymer, such as Al_2O_3 , ZnO , and Si_2O . The conductivity of the samples can keep low under 323 K and even show an obvious conductivity decrease at 333 K after the grafting, indicating the feasibility of the designed molecule in high voltage field. We observe a common positive effect of the stabilizers on the different electrical properties and find a similarity of the mechanism (Figure 7). The optimal molecules are advantageous for all the three electrical properties: higher TIV, lower conductivity, and better space charge inhibition. With regard to the similar mechanism, for electrical tree suppressing, the designed structure is like a micro electric field, in which hot electrons can be captured and oscillated. For space charge accumulation and conductivity decreasing, the aromatic nucleus is the bound area and the electron withdrawing and donating groups can serve as electron and hole traps. In the structure, the carriers of positive and negative polarity will be captured and even neutralized. Consequently, the same capture mechanism exists between electrical tree suppression and space charge inhibition as well as conductivity decreasing; meanwhile, the same trap mechanism exists between space charge inhibition and conductivity decreasing.

Nevertheless, some issues still need to be solved. For instance, the instability of amino may affect the

performance of the voltage stabilizers at high temperature. Therefore, we propose three strategies: (1) searching for another alternative group of amino; (2) protecting the amino group, that is, replacing the hydrogen atom by methyl; and (3) grafting the molecules to the PE chains through amino. The feasibility of strategy (2) and (3) has been demonstrated in the study as the molecule 3 adding and grafting results show. We expect to provide a new look for the design of voltage stabilizers through our research.

5 | CONCLUSIONS

In summary, all the delocalized aromatic molecules display enhanced insulation properties of polyethylene blends except for 4-cyanophenylboronic acid only containing electron-withdrawing groups. Comprehensively, 3-aminobenzoic acid showed the best electrical properties including great TIV increase, evident space charge inhibiting, and favorable conductivity decrease at lower temperature. The superior performance can be ascribed to the closest matching ability of electron-donating and electron-withdrawing between the amino and carboxyl. What is more, the intermolecular hydrogen bond may be another contributing factor. The results also coincide with DFT calculations, which show that significantly negative correlation trend exists between E_g and TIV and 3-aminobenzoic acid molecule has the lowest E_g . Besides, the grafting feasibility of the optimal molecule has been proved. Last but not least, the common capture and trap mechanism is emphasized in the effect of voltage stabilizers on the three different electrical properties.

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